

1. What is a computer network?

A computer network is a digital telecommunication network which allows different devices to connect to each other and share information.

2. What is the difference between a client and a server?

A client is a device that accesses the information from the server by sending the request to the server;

The server provides service, information, responses to the client based on the client's request.

3. What is a LAN?

A local area network is a type of network that lets devices connect to each other in a small area such as a home or office, etc.

4. What is a WAN?

A wide area network is a type of network that connects networks over large geographical areas such as cities, countries, etc.

5. What is the purpose of a switch?

It connects devices inside the LAN, it allows multiple devices to communicate with each other and send data where it needs to go, devices like computers, servers, etc.

6. What is the purpose of a router?

Connects different networks and lets them communicate with each other. It connects the home network with the internet.

7. What is a firewall?

The firewall monitors and controls network traffic according to security rules, it allows legitimate networks and blocks suspicious networks.

8. What is an IP address?

An IP address is the type of device's address on IP network, it can be IPv4 or IPv6.

9. What is a MAC address?

A device's physical address used on a network interface.

10. What is a default gateway?

The router or device sends data to when the data needs to be send outside the local network.

11. What is DNS?

Domain Name Server, It gives an IP address a domain name. It's like a phone book of name and numbers, where the name is the domain name and the number is the IP address.

12. Why is networking knowledge important for cybersecurity?

Networking is the prerequisite for tools like Wireshark, Burpsuite, Nmap, etc., as the security professionals constantly analyze the IP address, ports, protocols, packets, and connections.

13. Write your computer's private IPv4 address.

10.204.35.85

14. Write your default gateway.

10.204.35.18

15. Write your DNS server.

nslookup google.com

Server: 10.204.35.18

SCREENSHOTS (3 marks)

Attach screenshots of: ipconfig, ping, tracert

```
venom@VENOM:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 364 bytes 32798 (32.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 364 bytes 32798 (32.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.204.35.85 netmask 255.255.255.0 broadcast 10.204.35.255
    inet6 2401:4900:87a7:eccd:9f8d:2537:ca43:1ba3 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::83a4:13e:661d:1ec prefixlen 64 scopeid 0x20<link>
    ether 10:f6:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 99946 bytes 89963337 (85.7 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 46814 bytes 16243311 (15.4 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
venom@VENOM:~$ ping google.com
PING google.com (2404:6800:4013:802::66) 56 data bytes
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=1 ttl=113 time=199 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=2 ttl=113 time=222 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=3 ttl=113 time=243 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=4 ttl=113 time=61.6 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=5 ttl=113 time=83.6 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=6 ttl=113 time=106 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=7 ttl=113 time=129 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=8 ttl=113 time=28.0 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=9 ttl=113 time=175 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=10 ttl=113 time=196 ms
64 bytes from nv-in-f102.1e100.net (2404:6800:4013:802::66): icmp_seq=11 ttl=113 time=219 ms
^C
--- google.com ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10014ms
```

```
venom@VENOM:~$ traceroute google.com
traceroute to google.com (2404:6800:4013:802::65), 30 hops max, 80 byte packets
 1  2401:4900:87a7:eccd::5a (2401:4900:87a7:eccd::5a) 3.545 ms 3.513 ms 3.507 ms
 2  * * *
 3  fc00:0:2:74::1 (fc00:0:2:74::1) 204.733 ms * *
 4  * * *
 5  2404:a800:1a00:805::135 (2404:a800:1a00:805::135) 204.668 ms 2404:a800:1a00:805::131 (2404:a800:1a00:805::131) 204.663 ms 204.657 ms
 6  * * *
 7  2001:4860:1:1::14d4 (2001:4860:1:1::14d4) 204.357 ms 204.333 ms 204.327 ms
 8  2001:4860:0:1::78d3 (2001:4860:0:1::78d3) 204.322 ms 2001:4860:0:1::7819 (2001:4860:0:1::7819) 204.317 ms 205.205 ms
 9  2001:4860:0:1::78c0 (2001:4860:0:1::78c0) 205.188 ms 2001:4860:0:1::77d8 (2001:4860:0:1::77d8) 205.182 ms 2001:4860:0:1::78be (2001:4860:0:1::78be) 205.176 ms
10  2001:4860:0:1::7841 (2001:4860:0:1::7841) 205.171 ms 2001:4860:0:1::77c7 (2001:4860:0:1::77c7) 168.528 ms 2001:4860:0:1::779f (2001:4860:0:1::779f) 168.403 ms
11  2001:4860:0:1::4b88 (2001:4860:0:1::4b88) 168.396 ms 168.388 ms 2001:4860:0:1::4b8a (2001:4860:0:1::4b8a) 168.374 ms
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * nv-in-f101.1e100.net (2404:6800:4013:802::65) 153.520 ms 153.515 ms
```